Day 2

From Genomes to Surveillance Action

Genomic Surveillance of Surgical Site Infections & AMR
08:30–17:00

Course files: github.com/aryakaul/ughe-ssi-genomics-course

Day 2 · Morning

Time	Session
08:30	Recap: your questions from yesterday
08:50	**Lab 4:** Who is it, and what does it resist?
10:30	*Break*
10:45	**Lecture 4:** Typing, trees and outbreaks
11:30	**Lab 5:** Pathogenwatch
12:30	*Lunch*

Day 2 · Afternoon

Time	Session
13:15	**Lab 6:** Case study, a cluster on the surgical ward
15:00	*Break*
15:15	**Lecture 5:** Building a sustainable programme
16:00	Action planning
16:40	Knowledge check, evaluation, certificates

Your questions from yesterday

Answers to the exit tickets

> Today's files: github.com/aryakaul/ughe-ssi-genomics-course
> Lab 5–6 data are in `labs/data/`

Lab 4

Who Is It, and What Does It Resist?

Day 2 · 08:50–10:30 · 100 min
Pairs · swap driver and navigator at Part D

LO4: detect AMR genes and plasmids, type the isolate, and interpret the results alongside phenotypic AST

Open your handout

Lab 4 handout

`Lab4_species_typing_amr.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

The scenario

- A patient develops a **deep SSI 9 days** after an open fracture repair
- The lab reports *S. aureus*, **methicillin-resistant**
- The isolate was sequenced: you have its **assembly** (FASTA)

KEY MESSAGE

What can the genome add?

Teaching scenario. The genome is real public data from the Galaxy Training Network.

Plan for this lab

Part	What	Time
A	Get the genome	5 min
B	Who is it? Species and strain type	20 min
C	What does it resist? AMR genes	35 min
D	Is resistance mobile? Plasmids	15 min
E	Does it match the lab’s AST?	20 min

Get the genome

1. New history: `Lab4 – AMR`
2. **Upload** → **Paste/Fetch data** → paste the URL **from your handout**
3. **Type:** `fasta` · **Name:** `MRSA_contigs.fasta`

> **OPTIONAL**
>
> Your own Lab 2 assembly finished? Run every step on it too.

MLST: the strain type

Tool **MLST** → `MRSA_contigs.fasta` → scheme **automatic** → **Run**

```
MRSA_contigs.fasta   saureus   ST?   arcC(?) aroE(?) …
```

CHECK YOURSELF

1. Which **scheme** did MLST choose? What does it tell you?
2. What is the **ST**? Is it a well-known MRSA clone?

PART B2 · OPTIONAL

Species check with Kraken2

- Tool **Kraken2** on `MRSA_contigs.fasta`, with a **report**
- What is the top species?

> **KEY MESSAGE**
>
> **If the organism is wrong or the culture is mixed, every result after this is wrong.**

AMRFinderPlus (NCBI)

- **Input type:** nucleotide → `MRSA_contigs.fasta`
- **Organism:** `Staphylococcus_aureus` (finds point mutations too)
- Everything else: defaults → **Run** → open the **report**

KEY COLUMNS

Element symbol (gene) · **Class** (drug class) · **% Identity** · **% Coverage**

PART C2

staramr

- ResFinder + PointFinder + PlasmidFinder + MLST in one tool
- Input `MRSA_contigs.fasta` → defaults → **Run**
- Open **summary.tsv** and **detailed_summary.tsv**

PART C3

Record your findings

Gene / mutation	Drug class	AMRFinderPlus?	staramr?	On a plasmid?

Fill in the full table in your handout. The last column comes in Part D.

PART C

Check yourself: AMR genes

CHECK YOURSELF

1. Which gene explains the **methicillin resistance**? *(Glossary cheat sheet!)*
2. Did both tools find the same genes? If not, why might they differ?
3. What is staramr's **predicted phenotype**?

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

Is resistance mobile? Plasmids

1. In `detailed_summary.tsv`, find the **plasmid** rows (`rep…`) and note their **contig**
2. Mark resistance genes on the **same contig** as a plasmid

> **ALTERNATIVE**
>
> **ABRicate** with the **plasmidfinder** database

PART D

Check yourself: plasmids

CHECK YOURSELF

1. How many plasmid replicons? On which contigs?
2. Which resistance genes are probably on a **plasmid**? Why does that matter for infection control?
3. Why can’t we be **certain** from an Illumina assembly?

Does the genome match the lab's AST?

- Use the **AST report in your handout** (Part E)
- **It is hypothetical:** written for teaching, not the real AST for this genome
- For each antibiotic: which gene explains it? Do genome and AST **agree**?

PART E

Check yourself: genome vs AST

CHECK YOURSELF

1. Where do genome and AST **disagree**? Give two possible explanations for each.
2. What would you **tell the clinician**? Write one plain-language sentence.
3. Would you use the genome **instead of** AST to treat this patient?

What each tool told us

Tool	Answers
MLST	**Who?** Strain type, known clones
AMRFinderPlus / staramr	**What resistance?** Genes and mutations
PlasmidFinder	**Is it mobile?**
Comparing with AST	**How confident are we?**

Next: what about **many** isolates? → Lecture 4 and Lab 5

Extension for fast pairs

- **ABRicate** with the **card** and **vfdb** (virulence) databases
- Annotate the genome with **Bakta**
- Upload the FASTA to **ResFinder** on the web and compare

Adapted from the GTN tutorial “Identification of AMR genes in an assembled bacterial genome” (CC-BY 4.0).

Break

Back at 10:45 for Lecture 4

Lecture 4

Typing, Trees and Outbreaks

Day 2 · 10:45–11:30 · 45 min

LO5: type isolates and interpret a phylogeny to support or rule out an outbreak

“We’ve had 5 *Klebsiella* SSIs on Ward B in 6 weeks. Is this an outbreak?”

What is an outbreak?

- An outbreak means **transmission**: the same strain passing between patients
- Directly, or through staff, equipment or the environment

KEY MESSAGE

Same species ≠ same strain.
Same AST pattern ≠ same strain.

Typing: naming bacteria at different resolutions

Method	Like a…	Used for
Species	country	Clinical ID
MLST (7 genes) → ST	city	Is it a known high-risk clone?
cgMLST (1,000–3,000 genes)	street	Clusters within an ST
SNPs (whole genome)	house number	Outbreaks and transmission

MLST in one slide

- Seven “housekeeping” genes; each version gets a number
- The combination of 7 numbers = the **Sequence Type (ST)**
- Known high-risk clones: *K. pneumoniae* ST258, ST307 · MRSA ST8, ST22
- Tools do it automatically (Lab 4, Lab 5)

KEY MESSAGE

Same ST = possibly related. Different ST = not a recent outbreak.

SNP distances

- Align genomes to a reference
- Count the positions where they differ
- **Few SNPs** → recent common ancestor → possible transmission
- **Many SNPs** → not recent transmission

ROUGH GUIDE (VARIES BY STUDY)

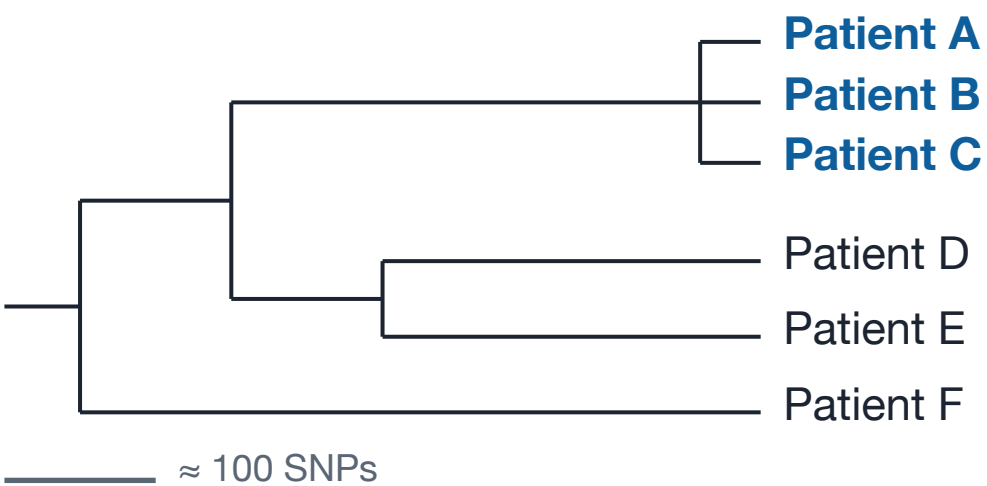
S. aureus: **≤ 15–30 SNPs**

K. pneumoniae, *E. coli*: **≤ 10–25 SNPs**

KEY MESSAGE

Thresholds are guides, not laws. Always combine them with epidemiology.

Reading a phylogenetic tree



- **Tips** = isolates
- **Branch length** = genetic change
- Tips on **short branches** close together are closely related (A, B, C)

COMMON MISTAKES

- Reading top-to-bottom order as meaningful
- Ignoring the **scale bar**

Genomics + epidemiology: time, place, person

A genomic cluster is an outbreak only if the **epidemiology fits**: same ward or theatre at overlapping times? Same surgeon, equipment or staff?

Confirm
Same strain, few SNPs, overlapping stays

Refute
Different STs: no need to close the ward

Find hidden links
Cases on different wards linked by shared equipment or staff

Strain outbreak vs plasmid outbreak

STRAIN OUTBREAK

The **same clone** in many patients

Tree: one tight cluster

PLASMID OUTBREAK

Different strains or species, but the **same resistance plasmid**

Needs Nanopore or hybrid data to confirm

KEY MESSAGE

Tree says “unrelated” but the resistance is unusual and identical? Think plasmid.

Vote: outbreak or not? (cards A / B / C)

1. **4 MRSA**, all ST22, 2–5 SNPs apart, same ward over 3 weeks
2. **4 *Klebsiella***: ST15, ST307, ST14, ST101
3. **3 *E. coli***, all ST131, 80–150 SNPs apart, 3 different hospitals

Vote: answers

1. **Likely outbreak**
2. **Not a single-strain outbreak.** Check for a shared plasmid?
3. **A common circulating clone**, not recent transmission

From result to action

Finding	Possible IPC action
Confirmed cluster	Enhanced cleaning, cohorting, contact screening, theatre audit, staff screening if indicated
Refuted cluster	Avoid unnecessary ward closure; look at other causes (e.g. prophylaxis timing)
High-risk clone or new carbapenemase	Notify the national reference lab / public-health authority

> Communicate in **plain language** to clinicians and managers. You'll practise this in Lab 6.

Tools for the next lab

PATHOGENWATCH: RESULTS

Upload assemblies → species, ST, AMR, clustering, tree

PATHOGENWATCH: CONTEXT

Add a metadata table → tree + timeline + metadata in one interactive view

Free, used by public-health labs, and needs **no coding**.

Lab 5

Pathogenwatch: Many Genomes at Once

Day 2 · 11:30–12:30 · 60 min
Pairs

LO5: use a free web platform to type, screen and visualise a collection of genomes

Open your handout

Lab 5 handout

`Lab5_pathogenwatch.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

The scenario

- The IPC nurse at **District Hospital A** is worried about more **MRSA SSIs on Surgical Ward B**
- The lab stored the isolates and sent them for sequencing
- You have **genomes from all MRSA isolates of the past months**

KEY MESSAGE

Get a quick overview of all the genomes at once.

The genomes are real public data. The hospital, wards and dates are fictional.

Plan for this lab

Part	What	Time
A	Upload: typing and AMR	30 min
B	Tree + timeline + metadata	25 min
	Summary	5 min

Files from the instructor

File	Contains
`assemblies/`	One FASTA per isolate
`lab6_metadata.csv`	Patient, ward, procedure and date information
`koser_snp_matrix.tsv`	SNP distances (used in Lab 6)

- **GitHub:** github.com/aryakaul/ughe-ssi-genomics-course
- Green **Code** button → **Download ZIP** → unzip → `labs/data/`

Upload and explore

1. Log in to `pathogen.watch` → **Upload**
2. Drag in the FASTA assemblies **and** `lab6_metadata.csv` **together**
3. For **at least 6** genomes, record:

Genome	Species	ST	Predicted resistances	QC pass?

Does the organism agree with the lab? Check the assembly QC metrics (length, contigs, N50).

Make a collection and view the tree

1. Select all your genomes → **Create collection** → name it `UGHE Lab 5`
2. Open the collection → explore the **tree** view
3. Explore the **AMR** view: resistance profiles side by side

CHECK YOURSELF

1. How many different STs are in the collection?
2. Do all isolates have the **same** AMR profile? What does that suggest, and what does it **not** prove?
3. In the tree, can you see a group of isolates on very **short** branches?

Find your metadata

1. Open the `UGHE Lab 5` collection
2. Look at the **data tables** at the bottom of the page
3. Find the columns `ward`, `theatre` and `hospital`

> No metadata columns? Raise your hand.

Explore: tree + time + place

- **Colour** the tree: click the `ward` column header, then `theatre`, then `hospital`
- Open the **Timeline**: when did each infection happen?
- **Click a tip**: find it in the table and timeline

CHECK YOURSELF

1. Is there a group that is closely related **and** close in time **and** in the same place?
2. Is any isolate from the **same ward at the same time not** in that group?
3. Is any isolate in the group from a **different ward**? What could link it?

PART B3

Save, share, summarise

Copy the link to your collection from the address bar. **You'll use it in Lab 6.**

Pathogenwatch gives you	Input
Species, ST, AMR, QC, clustering, global context	FASTA assemblies
A visual story: tree + time + place + metadata	Assemblies + a table (CSV)

Free and used by public-health labs. **You can use it on Monday with your own data.**

Pathogenwatch: Centre for Genomic Pathogen Surveillance.

Lunch

Back at 13:15 for the case study (Lab 6)

Lab 6

Case Study:
A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min
Groups of 4

LO3–LO5: combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

Open your handout

Lab 6 handout

`Lab6_ssi_outbreak_case_study.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

About this case

- The **genomes are real**: MRSA isolates from a published outbreak investigation
- The **hospitals, wards, patients and dates are fictional**, so the case feels like our setting

> The real story is revealed at the end.

THE SITUATION

District Hospital A · Monday, 16 March 2026

The IPC nurse writes to you:

> *“Since early February we’ve had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it’s bad luck. Theatre staff say it’s the ward. The ward says it’s the theatre.*

The question

> *The director asks whether we should **close Ward B**. That would cancel about 40 operations a week. We got genomes back last week for all MRSA isolates since September.*
>
> ***Can you tell us what's going on by Wednesday's IPC meeting?"***

Your evidence

1. **Pathogenwatch collection** from Lab 5: tree + timeline + metadata
2. **SNP matrix:** `koser_snp_matrix.tsv` (opens in Excel)
3. **Pathogenwatch results** from Lab 5: ST and AMR
4. **Metadata:** `lab6_metadata.csv`

Group roles

Role	Job
Genomics lead	Reads the tree and SNP matrix
Epidemiologist	Builds the timeline: who was where, when
Clinician / IPC	Thinks through transmission routes and actions
Communicator	Writes and gives the 5-minute briefing

Plan for this lab

Part	What	Time
A	Describe the cases	15 min
B	What does the genome say?	25 min
C	Genomics + epidemiology together	20 min
D	Prepare the briefing	25 min
E	Briefings to the IPC committee	15 min

Describe the cases

Sketch an **epidemic curve**: weeks across, number of cases up.
Two colours: **Ward B** and **other wards / hospitals**.

QUESTIONS

1. How many MRSA isolates in total? How many from Ward B since 1 February?
2. Do all Ward B cases look the same on the **lab report**? Does that make them one outbreak?

What does the genome say?

1. **Sequence types:** group the isolates by ST (Pathogenwatch)
2. **SNP distances:** highlight all pairs **≤ 30 SNPs** apart
3. Which isolates form a **cluster**, all close to each other?
4. **Tree:** colour by **ward**, then by **theatre**. Is the cluster a tight group on short branches?

≤ 30 SNPs: our working threshold for possible recent *S. aureus* transmission. Published thresholds vary.

Put genomics and epidemiology together

For **each** Ward B case since February:

Patient	In the cluster?	Ward / theatre / time link?	Linked · not linked · uncertain

Use the full table in your handout.

PART C

Discuss in your group

1. Is there an outbreak? How confident are you?
2. Is a Ward B patient **outside** the cluster, though on the ward at the same time?
3. Is a cluster patient **not** on Ward B? What do they share? → **likely route?**
4. Are isolates from **other hospitals** in the cluster?
5. What **other information** would you want?

Prepare a 5-minute briefing

For the IPC committee: director, chief surgeon, nursing lead, lab head. **Not genomics experts.**

RULES

- One flip-chart page or 3 slides
- Explain any jargon: “SNP” → “genetic differences”
- **No more than one figure**

PART D

Briefing template

1. **Bottom line:** “We found / did not find evidence of an outbreak of ___ linked to ____.”
2. **What we did:** “We sequenced __ MRSA isolates from __ to __.”
3. **What we found:** one visual + 2–3 bullets
4. **What it means:** which cases are linked; the likely route
5. **Recommendations:** 3 actions, with who and by when
6. **What we don’t know yet**

Briefings to the IPC committee

Each group has **5 minutes**. The committee must ask at least one question:

“Should we close Ward B?”

“Is it the surgeons' fault?”

“How sure are you?”

“What will this cost?”

Pause

Instructor: don’t go to the next slide until every group has given its briefing

The reveal

The genomes come from **Köser et al., *NEJM* 2012**: a suspected MRSA outbreak in a **neonatal unit** in Cambridge, UK.

- Sequencing separated the **outbreak isolates** from unrelated MRSA that looked identical on routine tests
- A follow-up study (**Harris et al. 2013**) traced transmission into the community and found a **staff carrier**, which ended the outbreak

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.

Take-home messages

1. **Same species + same antibiogram ≠ same outbreak**
2. Genomics finds **hidden links**: shared theatre, staff, equipment
3. Genomics can **prevent unnecessary actions**, like closing a ward
4. Genomics is only useful with **good metadata** and **clear communication**

Break

Back at 15:15 for Lecture 5

Lecture 5

Building a Sustainable Genomic Surveillance Programme

Day 2 · 15:15–16:00 · 45 min
≈ 25 min talk + 20 min discussion

LO6: design a sustainable, outsourced genomic surveillance workflow for your own setting

What we can do on Monday

Collect → **Store** → **Send** → **Analyse** → **Act**

No sequencer is needed to start. We need:

- a **plan** and a clear question
- **stored isolates** with good **metadata**
- a sequencing **partner**
- **people** who can analyse the data

Choose a question first

Type	Example	What to sequence
Outbreak (reactive)	“Is this ward cluster one strain?”	All cluster isolates + a few unrelated controls
Routine (proactive)	“Which ESBL/ carbapenemase genes cause our SSIs?”	e.g. all resistant SSI isolates for 6 months
Research	“Are post-caesarean SSI *E. coli* from the community?”	A designed sample with matched community isolates

Start small.

50–100 well-chosen isolates with complete metadata beat 1,000 random ones.

Sampling and biobanking

- Store **every** SSI isolate with resistance of interest (at least ESBL, carbapenem-resistant, MRSA)
- –80 °C freezer or glycerol stocks; a **backup** at a second site if possible
- Keep an **isolate register** (spreadsheet): study ID, organism, date, ward, specimen, AST, freezer location
- Agree on labelling **before** you start

Metadata and data management

- Use a template aligned with **PHA4GE** fields
- **Coded IDs only.** Keep the ID → patient key secure and separate
- Name files consistently: `UGHE-SSI-0001_R1.fastq.gz`
- Back up raw reads in **two places**: they can't be replaced
- Galaxy histories are a free, reproducible record of every analysis

Ethics, agreements and data sharing

- **Surveillance and IPC:** usually covered by routine public-health practice
- **Research:** needs ethics approval (Rwanda National Ethics Committee + institutional IRB)
- **Sending isolates abroad:** material transfer agreement (MTA)
- **Data-sharing agreement:** who owns the data and how it is published
- **Share genomes publicly** (ENA / NCBI) with non-identifying metadata

KEY MESSAGE

Equity: local investigators should lead the analysis and authorship.

Partners in the region

RBC
Rwanda Biomedical Centre / National Reference Laboratory

Africa CDC PGI
Regional sequencing and training network

ASLM
AMR surveillance programmes such as MAAP

Other partners
Universities, NGOs, Fleming Fund projects, commercial services

Ask the room: which of these do we already work with?

Reporting results so they change practice

Audience	What they need
Clinicians	“This isolate carries an NDM carbapenemase. Avoid carbapenems; consult ID.”
IPC committee	Cluster summary, tree, timeline, recommended actions
Ministry / RBC	Quarterly summary of resistance genes and clones

> Use simple visuals (Pathogenwatch) and a one-page template. **Close the loop:** did the action reduce SSIs?

People: keeping the skills alive

- **2–3 “genomics champions”** per department
- A monthly **data clinic / journal club** for course graduates
- Self-paced learning: Galaxy Training Network, The Carpentries, Africa CDC, Wellcome Connecting Science
- Build genomics into **student teaching** (MGHD, MBBS research projects)

Funding

- Build sequencing costs into grants (AMR, surgery, maternal health)
- Funders active in Africa: Fleming Fund, Wellcome, Gates Foundation, Africa CDC, NIH / Fogarty, EDCTP

BUDGET PER ISOLATE

Sequencing + shipping + staff time + data storage

Your one-page action plan

1. **Question:** what do I want to know?
2. **Isolates:** which, how many, from where, when? Already stored?
3. **Metadata:** which fields? Who collects them?
4. **Partner:** who? Which platform? Cost? Timeline?
5. **Analysis:** which tools? Who does it?
6. **Ethics:** IRB? MTA? Data sharing?
7. **Communication:** who gets results, in what format?
8. **First step next week:** _____

Discussion

20 min · small groups, then the whole room

1. What is the **single biggest barrier** to starting genomic surveillance at our institution? How could we overcome it?
2. Which **question** (outbreak, routine, research) would be most valuable to answer first?

Action planning

Draft **your own one-page plan** for a genomic surveillance mini-project.

- Use the 8-point template from Lecture 5
- Choose a question you could really start on next week
- Be concrete: **which** isolates, **which** partner, **who** does what

KEY MESSAGE

Last point on the template: your first step next week.

Knowledge check: after the course

- The **same questions** as yesterday
- Use the **same code word** as yesterday, and write **Post**
- Afterwards we'll go through the answers together

Knowledge check · Post

- **Not graded**, and anonymous
- On your quiz sheet or a piece of paper, write your **code word** and **Post**
- For each question, write its number and your answer (e.g. “1 b”)
- One question at a time on the screen

KEY MESSAGE

It’s fine not to know an answer. Just leave it blank.

KNOWLEDGE CHECK · QUESTION 1 OF 15

Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

KNOWLEDGE CHECK · QUESTION 2 OF 15

Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 3 OF 15

Name two bacterial species that commonly cause surgical site infections.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 4 OF 15

An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads

c) AST

d) MLST

KNOWLEDGE CHECK · QUESTION 6 OF 15

A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30×

c) 150×

d) 300×

What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores

d) Gene names

After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50

c) A total length twice the expected genome size

d) No contigs

The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance)

c) Carbapenem resistance

d) A virulence toxin

blaNDM-1 is an example of:

a) An ESBL

b) A carbapenemase

c) An efflux pump

d) A housekeeping gene

The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

Write your answer on your paper

Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types

c) Only if they are both ESBL

On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them

c) The colour of the labels

Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak

c) Contamination of the sequencer

KNOWLEDGE CHECK · QUESTION 15 OF 15

List three pieces of metadata that must be recorded with every isolate for surveillance.

Write your answer on your paper

KNOWLEDGE CHECK

How confident are you?

Rate each from **1** (not at all) to **5** (very confident):

- I can explain what a genome sequence can tell us about an SSI
- I can run a tool in Galaxy
- I can identify resistance genes from a genome
- I can interpret a phylogenetic tree
- I can plan a genomic surveillance project

N4

What further training or support would help you most after this course?

Let’s go through the answers

Hand in your paper first, then follow along

Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain ✓

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

WHY

Culture and AST guide treatment for this patient. Only the genome shows whether isolates are the **same strain**.

Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

ANSWER

No. Isolates or DNA can be sent to a partner. The local skills are choosing isolates, recording metadata, analysing and acting on results.

Name two bacterial species that commonly cause surgical site infections.

ANSWER

Any two of: *S. aureus*, *E. coli*, *K. pneumoniae*, *P. aeruginosa*, *A. baumannii*, *Enterobacter*, *Enterococcus*.

An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate ✓

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

WHY

Illumina gives short, very accurate reads. Long reads come from Nanopore.

Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads ✓

c) AST

d) MLST

WHY

Long reads can span a whole plasmid, so you can see which genes sit on it.

A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30× ✓

c) 150×

d) 300×

WHY

1,000,000 × 150 ÷ 5,000,000 = **30×**

What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores ✓

d) Gene names

WHY

FASTQ = sequence + a **quality** line for every read. FASTA = sequence only.

After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50 ✓

c) A total length twice the expected genome size

d) No contigs

WHY

Fewer, longer pieces. A total length twice the expected size suggests contamination.

The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance) ✓

c) Carbapenem resistance

d) A virulence toxin

WHY

mecA makes an altered target (PBP2a), so almost all β-lactams stop working.

`blaNDM-1` is an example of:

a) An ESBL

b) A carbapenemase ✓

c) An efflux pump

d) A housekeeping gene

WHY

NDM = New Delhi metallo-β-lactamase: it destroys carbapenems.

The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

ANSWER

Any of: an unknown mechanism · a mutation not in the database · low coverage missed the gene · mixed culture · lab error.

Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types ✓

c) Only if they are both ESBL

WHY

Different STs are not one recent transmission chain. (They could still share a plasmid.)

On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them ✓

c) The colour of the labels

WHY

Branch length = genetic change. Top-to-bottom order means nothing: branches can rotate.

Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak ✓

c) Contamination of the sequencer

WHY

Very few SNPs **and** an epidemiological link: possible transmission.

List three pieces of metadata that must be recorded with every isolate for surveillance.

ANSWER

Any three of: collection date · facility · ward · specimen type · surgery type and date · AST results · organism · coded patient ID.

Course evaluation

- What worked well?
- What should we change for the next group?
- Anything still unclear?

KEY MESSAGE

Your answers shape the next version of this course.

Keep going

- **Course files:** github.com/aryakaul/ughe-ssi-genomics-course
- Your **Galaxy histories** stay in your account: re-run them on new data
- Next steps: `resources/further_learning.md`
- A monthly **data clinic / journal club** for course graduates

Thank you!

Certificates now